

Question ID: 585a9138

| Assessment | Test | Domain                    | Skill                        | Difficulty                                          |
|------------|------|---------------------------|------------------------------|-----------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Lines, angles, and triangles | Medium <div><div></div><div></div><div></div></div> |

Question

Triangles  $ABC$  and  $DEF$  are congruent, where  $A$  corresponds to  $D$ , and  $B$  and  $E$  are right angles. The measure of angle  $A$  is  $27^\circ$ . What is the measure, in degrees, of angle  $F$ ?

Correct Answer: 63

Rationale

The correct answer is **63**. It's given that triangles  $ABC$  and  $DEF$  are congruent with angle  $A$  corresponding to angle  $D$ . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle  $A$  is  $27^\circ$ . It follows that the measure of angle  $D$  is also  $27^\circ$ . It's given that angle  $E$  is a right angle. Therefore, the measure of angle  $E$  is  $90^\circ$ . Let  $x$  represent the measure, in degrees, of angle  $F$ . Since the measures of the angles in a triangle sum to  $180^\circ$ , it follows that  $27 + 90 + x = 180$ , or  $117 + x = 180$ . Subtracting **117** from both sides of this equation yields  $x = \mathbf{63}$ . Therefore, the measure, in degrees, of angle  $F$  is **63**.